

KGATLA MAHLATSE. NORMAN

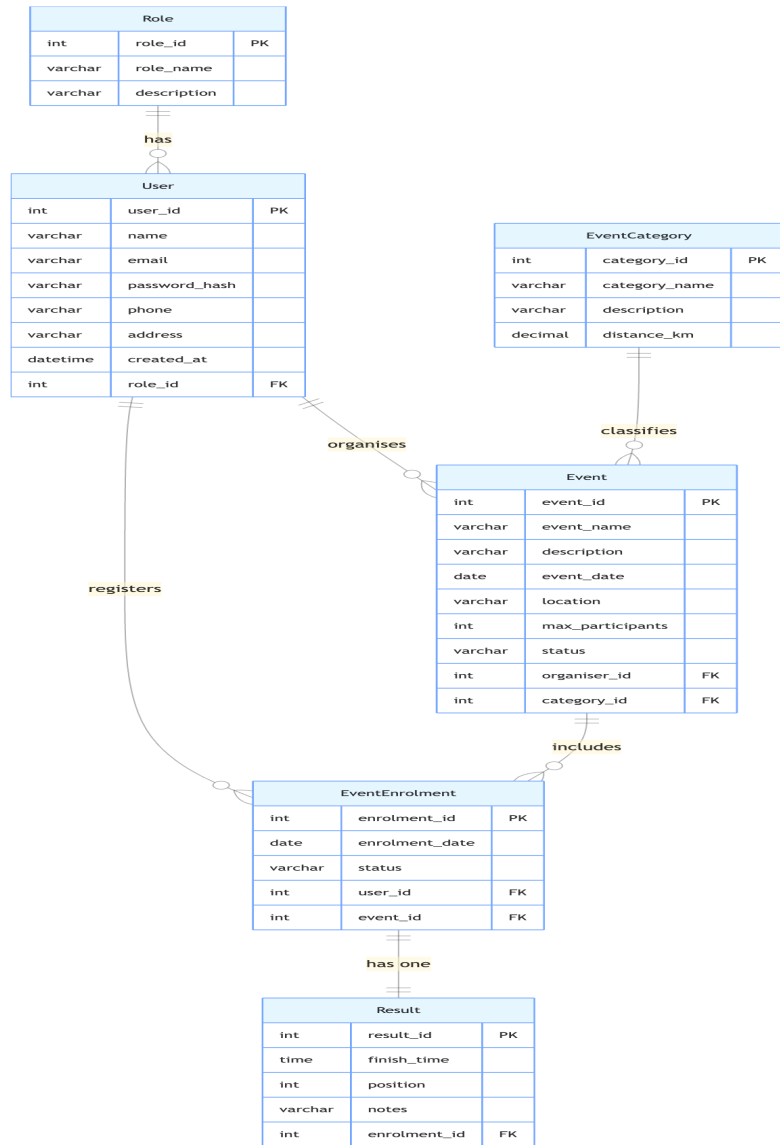
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DUE DATE: 04 AUG 2026

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SECTION A



SECTION B

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register new participant account	Public	{name, email, password, phone, address}	201 Created — user + token 400 Bad Request — validation errors 409 Conflict — duplicate email
POST	/api/auth/login	Authenticate & get JWT token	Public	{email, password}	200 OK — token + user info 401 Unauthorized — invalid credentials
GET	/api/users/profile	Get logged-in user's profile	Participant, Organiser, Admin	—	200 OK — user details 401 Unauthorized
PUT	/api/users/profile	Update own profile	Participant, Organiser, Admin	{name, phone, address}	200 OK — updated user 401 Unauthorized

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	List all upcoming events	Public	—	200 OK — event list (basic info)
GET	/api/events/{id}	Get single event details	Public	—	200 OK — full event details 404 Not Found
POST	/api/events	Create new event	Organiser, Admin	{event_name, description, event_date, location, max_participants, category_id}	201 Created — event 403 Forbidden 400 Bad Request
PUT	/api/events/{id}	Update event	Organiser (owner), Admin	{event_name, description, event_date, location, max_participants, status, category_id}	200 OK — updated event 403 Forbidden 404 Not Found
DELETE	/api/events/{id}	Delete/cancel event	Organiser (owner), Admin	—	204 No Content 403 Forbidden 404 Not Found
GET	/api/categories	List all event categories	Public	—	200 OK — category list
POST	/api/categories	Add new category	Organiser, Admin	{category_name, description, distance_km}	201 Created — category

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					409 Conflict — duplicate name
GET	/api/events/{eventId}/enrolments	View enrolments for an event	Organiser (owner), Admin	—	200 OK — enrolment list 403 Forbidden
POST	/api/events/{eventId}/enrolments	Enrol in event	Participant	—	201 Created — enrolment (Pending) 409 Conflict — already enrolled 403 Forbidden — event full/closed
PUT	/api/enrolments/{id}	Approve/reject enrolment	Organiser (owner), Admin	{status}	200 OK — updated enrolment 404 Not Found
GET	/api/users/enrolments	My enrolments	Participant	—	200 OK — user's enrolments list
POST	/api/enrolments/{id}/results	Submit race result	Organiser (owner), Admin	{finish_time, position, notes}	201 Created — result 409 Conflict — result exists

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
					404 Not Found
GET	/api/events/{eventId}/results	View all results for event	Public	—	200 OK — results with positions
GET	/api/users/results	My results	Participant	—	200 OK — personal results

SECTION C

```
-- =====  
  
-- RaceDay System — SSMS Database Script  
-- Script Name: part1_poe  
-- Creates schema + seed data  
-- Tested on: SQL Server 2016+  
-- =====  
  
-- Switch to master FIRST and STAY there until DROP/CREATE done  
USE master;  
  
-- Prevent "database in use" errors — kick all connections  
IF EXISTS (SELECT name FROM sys.databases WHERE name = N'part1_poe')  
BEGIN  
    ALTER DATABASE part1_poe SET SINGLE_USER WITH ROLLBACK IMMEDIATE;  
    ALTER DATABASE part1_poe SET MULTI_USER; -- release lock  
    DROP DATABASE part1_poe;  
END  
  
-- Create fresh database  
CREATE DATABASE part1_poe;  
  
-- EXPLICITLY switch — forces context (CRITICAL FIX)  
USE part1_poe;
```

```

-- =====
-- 1. Roles Table
-- =====
CREATE TABLE Role (
    role_id INT IDENTITY(1,1) PRIMARY KEY,
    role_name VARCHAR(50) NOT NULL UNIQUE,
    description VARCHAR(200)
);

-- =====
-- 2. Users Table
-- =====
CREATE TABLE [User] (
    user_id INT IDENTITY(1,1) PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    phone VARCHAR(20),
    address VARCHAR(255),
    created_at DATETIME DEFAULT GETDATE(),
    role_id INT NOT NULL,
    FOREIGN KEY (role_id) REFERENCES Role(role_id)
);

-- =====
-- 3. Event Categories Table
-- =====
CREATE TABLE EventCategory (
    category_id INT IDENTITY(1,1) PRIMARY KEY,
    category_name VARCHAR(100) NOT NULL,
    description VARCHAR(255),

```

```
distance_km DECIMAL(5,2) NOT NULL,  
CONSTRAINT UQ_CategoryName UNIQUE (category_name)  
);
```

```
-- =====
```

```
-- 4. Events Table
```

```
-- =====
```

```
CREATE TABLE [Event] (  
    event_id INT IDENTITY(1,1) PRIMARY KEY,  
    event_name VARCHAR(150) NOT NULL,  
    description TEXT,  
    event_date DATE NOT NULL,  
    location VARCHAR(150) NOT NULL,  
    max_participants INT NOT NULL CHECK (max_participants > 0),  
    status VARCHAR(20) DEFAULT 'Upcoming' CHECK (status IN ('Upcoming', 'Open', 'Closed', 'Completed', 'Cancelled')),  
    organiser_id INT NOT NULL,  
    category_id INT NOT NULL,  
    FOREIGN KEY (organiser_id) REFERENCES [User](user_id),  
    FOREIGN KEY (category_id) REFERENCES EventCategory(category_id)  
);
```

```
-- =====
```

```
-- 5. Event Enrolments Table
```

```
-- =====
```

```
CREATE TABLE EventEnrolment (  
    enrolment_id INT IDENTITY(1,1) PRIMARY KEY,  
    enrolment_date DATE DEFAULT CAST(GETDATE() AS DATE),  
    status VARCHAR(20) DEFAULT 'Pending' CHECK (status IN ('Pending', 'Approved', 'Rejected', 'Withdrawn')),  
    user_id INT NOT NULL,  
    event_id INT NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES [User](user_id) ON DELETE CASCADE,  
    FOREIGN KEY (event_id) REFERENCES [Event](event_id) ON DELETE CASCADE,
```

```
CONSTRAINT UQ_UserEvent UNIQUE (user_id, event_id)
);
```

```
-- =====
-- 6. Results Table
-- =====
```

```
CREATE TABLE Result (
    result_id INT IDENTITY(1,1) PRIMARY KEY,
    finish_time TIME NOT NULL,
    position INT CHECK (position > 0),
    notes VARCHAR(255),
    enrolment_id INT NOT NULL UNIQUE,
    FOREIGN KEY (enrolment_id) REFERENCES EventEnrolment(enrolment_id) ON DELETE CASCADE
);
```

```
-- =====
-- SEED DATA
-- =====
```

```
-- Roles
INSERT INTO Role (role_name, description) VALUES
('Admin', 'System administrator'),
('Organiser', 'Can create and manage events'),
('Participant', 'Can enrol and view results');
```

```
-- Users (IDs: 1=Sarah, 2=Mike, 3=Thabo, 4=Lerato)
INSERT INTO [User] (name, email, password_hash, phone, address, role_id) VALUES
('Sarah Johnson', 'sarah@racehub.com', 'HASH_ORG1', '011-555-0101', 'Johannesburg', 2),
('Mike Dube', 'mike@racehub.com', 'HASH_ORG2', '011-555-0102', 'Pretoria', 2),
('Thabo Mkhize', 'thabo@email.com', 'HASH_P1', '082-555-0103', 'Sandton', 3),
```

```
('Lerato Ndlovu', 'lerato@email.com', 'HASH_P2', '083-555-0104', 'Midrand', 3);
```

```
-- Categories (IDs: 1=5K, 2=10K, 3=Half Marathon, 4=Full Marathon)
```

```
INSERT INTO EventCategory (category_name, description, distance_km) VALUES
```

```
('5K Fun Run', 'Short distance recreational run', 5.00),
```

```
('10K Road Race', 'Standard 10km road event', 10.00),
```

```
('Half Marathon', '21.1km intermediate distance', 21.10),
```

```
('Full Marathon', '42.2km classic endurance race', 42.20);
```

```
-- Events
```

```
INSERT INTO [Event] (event_name, description, event_date, location, max_participants, status, organiser_id, category_id)  
VALUES
```

```
('Johannesburg Spring Run', 'Annual spring 10km through the city', '2026-10-15', 'Johannesburg Zoo', 500, 'Upcoming', 1, 2),
```

```
('Sandton 5K Twilight', 'Evening fun run with lights', '2026-11-05', 'Sandton City', 300, 'Open', 1, 1),
```

```
('Pretoria Half Marathon', 'Scenic route around the Union Buildings', '2026-12-01', 'Pretoria CBD', 800, 'Open', 2, 3);
```

```
-- Enrolments (Generated IDs: 1, 2, 3, 4)
```

```
INSERT INTO EventEnrolment (status, user_id, event_id) VALUES
```

```
('Approved', 3, 1), -- enrolment_id = 1
```

```
('Pending', 4, 1), -- enrolment_id = 2
```

```
('Approved', 3, 2), -- enrolment_id = 3
```

```
('Approved', 4, 3); -- enrolment_id = 4
```

```
-- Results  Uses enrolment IDs 1 and 3 — NO duplicates!
```

```
INSERT INTO Result (finish_time, position, notes, enrolment_id) VALUES
```

```
('00:48:32', 12, 'Personal best!', 1),
```

```
('00:35:20', 8, 'Great run!', 3);
```

```
PRINT  part1_poe created successfully with seed data.;
```

```
PRINT  All constraints passed — NO duplicate key errors!;
```

```
PRINT ";
```

```
-- =====
```

```
-- VERIFICATION
```

```
-- =====
```

```
SELECT
```

```
  'Role' AS TableName, COUNT(*) AS TotalRecords FROM Role UNION ALL
```

```
SELECT 'User', COUNT(*) FROM [User] UNION ALL
```

```
SELECT 'EventCategory', COUNT(*) FROM EventCategory UNION ALL
```

```
SELECT 'Event', COUNT(*) FROM [Event] UNION ALL
```

```
SELECT 'EventEnrolment', COUNT(*) FROM EventEnrolment UNION ALL
```

```
SELECT 'Result', COUNT(*) FROM Result;
```